Data in Motion: Designing Motion-to-Sound Within the Limits of Low-Cost Motion Tracking

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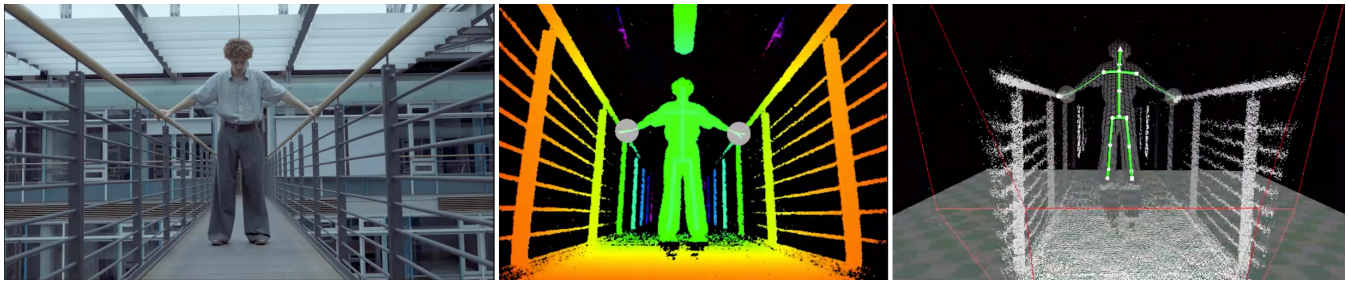


Figure 1: *Kinect V2 tracking — the distance between the performer's hands widens a panning sound effect (Scenario 1).*

Abstract

While professional motion tracking relies on expensive, complex infrastructure that remains out of reach for most independent artists, affordable sensors have democratized motion capture for embodied interaction design. Translating movement into an artistic interpretation, however, rests on a chain of design decisions bound by the data stream's reliability. How low-cost failures force such decisions is rarely documented. This paper examines a motion-to-sound pipeline through iterative testing across two use case scenarios with a performer, drawing on coded recordings and a written questionnaire. The sensor's constraints led to concrete choices: computing movement from relational body data rather than coordinates in space, setting aside unreliable parameters and gestures. Within them, the performer developed a movement vocabulary and described this relational account as matching his own movement perception. Our findings encourage artists to explore affordable motion tracking and its limits in an experimental way.

1. Introduction

The exploration of the body as an interface between the physical and digital worlds has gained growing popularity within a flourishing media art culture since advances in sensor technology [WNW24] — a new bridge between the digital and performative arts. To create an artistic emergence from this interaction with technology, the body's kinematic state must be captured and transmitted into a processable data stream. High-end professional tracking setups remain costly and resource-intensive: optical systems, for instance, require full-surround camera coverage and specially equipped spaces [MIGL17].

Several studies suggest that affordable alternatives to professional motion capture can lead to meaningful results [VPK*25, MIGL17]. Yet access to motion data is not the same as access to expression. There is a deeper level than data collection, as illustrated by the connection between movement and sound: between body data and auditory feedback lies a space for decision-making that must be designed. Mapping determines not only which parts of the body become audible, but also how they are oriented in relation to the space or the body.

How the tension between artistic intent and technical possibilities can shape creative processes has been explored [FA19]. Data processing using the body's own reference frame has also been applied in earlier projects [DSD19]. This case study follows the development of the mapping approach, examining how limitations of a low-cost motion-to-sound setup shaped design decisions and a performer's movement vocabulary. The work was situated within a collaborative performance project by actors and creative technology students. Within this context, We developed the motion-data processing and mapping approach examined in this paper. Its first and obvious approach, which mapped sound to absolute positions in the room, proved unstable and was replaced by relational motion

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data calculation. Existing video and audio materials from an exploratory session and a public live performance were coded event by event.

For media artists and readers working in embodied interaction design, this paper examines the data processing of a motion-to-sound pipeline (Sections 3.1–3.2) within the limitations of the system. Based on the typology of failures and observations, the analysis highlights key design decisions, such as the shift to relational body data (Section 3.4). Furthermore, it relates these findings to the performer's experiences as gathered from a written questionnaire (Sections 4.2–4.3) and aims to share the lessons learned in the context of collaborative work with performing artists.

2. Related Work

Low-cost motion capture. Barriers arising from high costs and complexity in motion capture motivated the development of alternatives: *Mesquite* [VPK*25], an open-source body-worn system, is reported to achieve accuracy comparable to the OptiTrack system at approximately 5% of the cost of the professional setup. Camera-based systems are also explored: When Microsoft released the Kinect Software Development Kit (SDK) for Windows in 2012, the sensor became usable on Windows machines, and projects in education, healthcare, robotics, entertainment and art were produced [Zha12]. Müller et al. [MIGL17] compared a gait analysis setup of six Kinect cameras with the professional Vicon system, with good agreement. Both studies, however, were conducted under controlled conditions and are difficult to apply to uncontrolled settings such as a stage. *So, how are low-cost systems used in live creative contexts?*

Interactive technology on stage. *.cyclic* uses Kinect technology to synthesize graphics, sound, and movement in a dance piece, documented by its authors [JJLM12]. Since skeleton tracking was unable to follow the dancers' movements specified in the choreography, it was discarded (one of the obstacles discussed in Section 3.3). *SenseStage* [BSdB*10] addresses the economic and technical constraints of live performance with a low-cost wireless sensor infrastructure, leaving short rehearsals free for artistic exploration. In both cases, the answer lies in the technology rather than in the work itself. Other work shifts the focus to the artistic process as a research methodology. Through her dance piece *SKIN*, Fdili Alaoui [FA19] explores the tensions between artistic intent and interactive technologies through a "research through practice" approach. The iterative process was used to identify possibilities and limitations in a delicate balancing act. Masu et al. [MCJ*19] developed design guidelines for dance productions from interviews with the dancers, describing interactive technology as a co-creator. Methodologically, this paper stands closest to Fdili Alaoui's approach. While embracing practice as a research method that includes the performer's perception, we reinforce our study with a coded analysis.

Mapping and constraints. In the connection between motion and sound, Hunt et al. [HWP02] show that altering the mapping, while leaving interface and sound source unchanged, changes the character of an instrument entirely. OtoKin [DSD19] applies this in dance improvisation: a many-to-many mapping of Kinect data makes small changes in posture audible and invites performers to re-learn position as sound. Schacher [Sch10] distinguishes the global stage space captured by camera tracking from a body-centric space providing detailed information within a limited scope. Magnusson [Mag10] treats constraints themselves as design material, arguing that limitations delimit a search space and engender creativity. This paper examines this in practice by tracing how concrete failures of one low-cost setup turned into design decisions and shaped a performer's movement vocabulary.

3. *Embodied Systems*: From Body Movement to Sound

We present *Embodied Systems*, which was developed across iterative studio sessions and translates body movement into corresponding audio output. The choice of components was shaped by factors typical of a student context: financial and organizational accessibility, mobility, interoperability, prior experience, and feasibility within the project's timeframe. When a person enters the tracking area, motion data is streamed in real time and alters a sound composition. The body's interaction with the system creates a synergy between the physical and virtual realm, while the body can create a narrative through physical expression and sound: a reciprocal dialogue is intended to emerge, following the notion of interactive technology as a co-creator (Section 2).

In Section 3.3 the trade-offs of this low-cost motion tracking pipeline identified during three development tests are described. The pipeline was then used in two Use Case Scenarios, an exploratory, open-ended session and a public live performance in two runs, which are analyzed in Section 4.

3.1. System Architecture

The technical setup was built from the following components: a Microsoft Kinect V2 depth camera for motion tracking, TouchDesigner for real-time data processing, and Ableton Live for sound synthesis. The setup uses two laptops connected through a wired network, since running TouchDesigner and the DAW together on a single laptop caused clipping problems due to CPU overload. Control data is exchanged between them via the Open Sound Control (OSC) protocol.

The Kinect V2 is no longer manufactured, which limits the reproducibility of this hardware setup [KHCC22]. TouchDesigner was selected because of prior experience with the software, Ableton Live because a university-provided license was available. This is relevant for reproducibility, as the pipeline is therefore not fully low-cost for artists working without institutional licenses.

3.2. Pipeline and Data Flow

The pipeline is divided into three main stages (Figure 2):

Motion Tracking. The Kinect sensor is connected via USB (Laptop 1) and provides real-time joint positions for up to 25 body joints that can be streamed in TouchDesigner or monitored in Kinect Studio. Each joint delivers absolute spatial coordinates.

Data Processing. TouchDesigner (Laptop 1 + 2) handles all data interpretation and motion-parameter extraction. The Kinect CHOP

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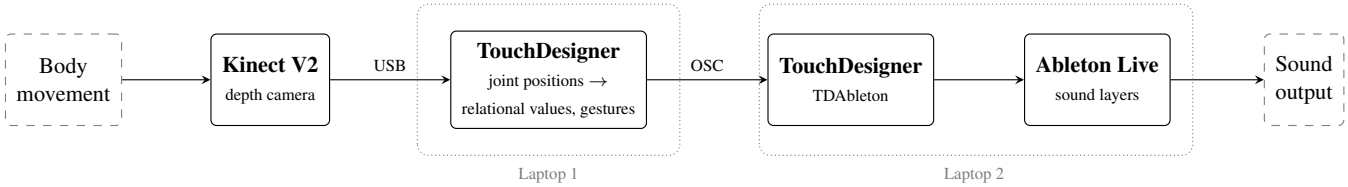


Figure 2: *Pipeline of Embodied Systems across two laptops.*

feeds the data stream of the individual joints into the node environment on Laptop 1. Raw joint positions are normalized and sent via OSC to another TouchDesigner file on Laptop 2. The integrated tool TDAbleton establishes a direct connection to Ableton Live.

Sound Synthesis. Ableton Live receives normalized motion parameters and maps them to different sound parameters, such as volume, expansion, or spatial panning (see Table 1). The sound tracks and triggers were assigned differently Scenario 1 and 2.

3.3. Trade-offs and Fail Cases

The pipeline's development presented various challenges. Test 1, an initial session with students of the acting class, served to become familiar with three motion values: body height (the head's vertical position relative to the camera's origin), the distance between the hands, and body speed (the slope of the body's center). Tests 2 and 3 took place with the performer in a theater hall (Figure 3): Test 2 examined a calibration process, Test 3 the relational body data (Section 3.4). In each test, the values were observed in TouchDesigner while the performer moved to basic sound material, and his feedback informed the next iteration (Figure 4).



Figure 3: *Test session with the performer: Laptop 1 (left) shows the skeleton tracking, Laptop 2 (right) runs Ableton Live.*

Limited camera perspective: A significant problem that comes with camera-based sensors is the limited camera range. Once the sensor is placed, it offers a reliable depth range of 0.5 to 4.5 meters within a field of view of 70.6° horizontally and 60° vertically [FBR*15]. Consequently, the sensor must be carefully positioned and movements planned and limited to the tracking area in advance.

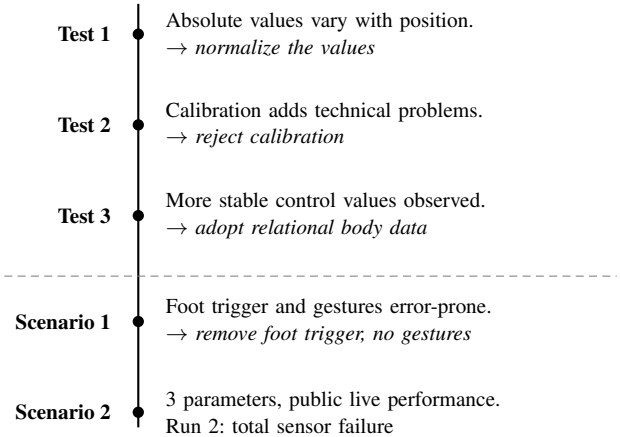


Figure 4: *Iterative pipeline development, including observed problems and resulting design decisions.*

Overlapping of joints: The overlapping of joints during certain movements, such as crossing legs or kneeling, causes the Kinect's tracking to falter (Figure 5): skeleton fitting is viewpoint-dependent, and the SDK infers the positions of occluded joints [MIGL17]. If these parts are mapped to sound parameters, there is a risk of confusion among the perceiving audience. This factor shaped the sound mapping: sounds were assigned mostly to upper body parts, and the foot speed trigger was excluded after Use Case Scenario 1. Kneeling positions and body turns were avoided in Scenario 2 and movements had to be oriented toward the camera.

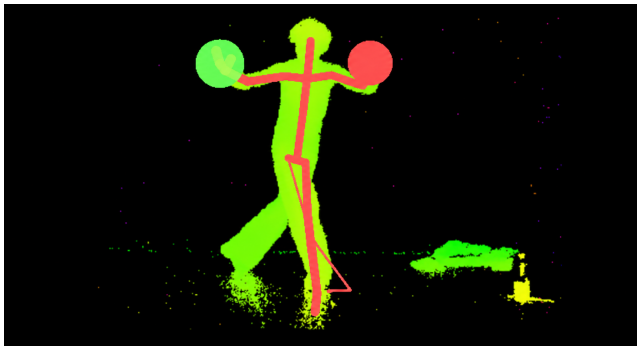


Figure 5: *Tracking failure during rehearsal: both leg joints collapse onto the standing leg instead of following the extended leg.*

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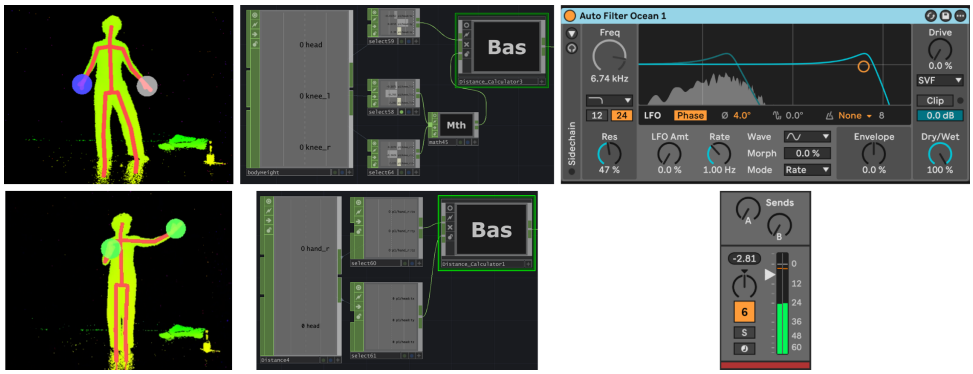


Figure 6: *Continuous mappings of Scenario 2 (rehearsal): tracking, calculation in TouchDesigner, effect in Ableton Live.*

Top: Body height → auto-filter. Bottom: Head–hand distance → heartbeat volume

Absolute data relations: A significant obstacle was the initial use of absolute spatial relationships (Tests 1 and 2). The Kinect SDK reports joint positions in *camera space*, a coordinate system whose origin lies at the center of the sensor's infrared camera, with values given in meters [Mic14]. Absolute coordinates depend on the body's position relative to the camera. Tracking accuracy also varies across the field of view, with depth errors increasing from about ±6 mm at the image center to roughly 30 mm near its edges at a distance of two meters [FBR*15]. Particularly in performance, where intuition and spontaneity come into play, the same movement would thus yield values depending on the position. As a first attempt to stabilize the motion data, a calibration process was implemented (Test 2): the coordinate range for selected joints was calculated and normalized within a ten-second window. The approach was rejected because instead of simplifying the setup it caused additional technical problems.

3.4. Solution: Relational Body Data

To reduce the mapping's dependence on the performer's position in space, we shifted to distances between body joints (Test 3). This changes the reference frame of the mapping but does not eliminate errors in the tracked joint positions. The spacing is calculated as the Euclidean distance between two joint positions:

$$d = \|p_a - p_b\|$$

The distance is computed with a freely available TouchDesigner component [Lee22] (Figure 6; e.g. head-to-hand distance) (see Table 1). This led to a conceptual shift: the body becomes a reference point and enters into an interaction with itself. Compared to the first two tests, more stable values were observed in Test 3. How the performer perceived this change is discussed in Section 4.3.

3.5. Gesture Detection

While other sensors already detect specific gestures, the Kinect V2 does not provide values for predefined poses. To interpret body poses, another component was added to the pipeline: a Gesture-Detection script. The script was executed via a CHOP Execute DAT in TouchDesigner. It uses joint positions and calculated distances from the relational data to determine poses in real time based on specific conditions (see Table 1).

4. Evaluation

4.1. Study Design

Given that an interactive system must exhibit a certain level of reliability, we examine it first: for a dialogue between the performer and the system to emerge at all, the movement-sound relationship must be coherent and synchronized. Two Use Case Scenarios are analyzed that built upon one another, both recorded on video and audio. All recordings were coded by the author and no second coder was involved. The evaluation draws on two sources: these recordings, compared in retrospect with the mappings in Table 1, and a written questionnaire on the performer's perception, which, as he took part in the development process, also bases on his subjective insight. The recordings are examined with two case-specific procedures:

Scenario 1, an exploratory, unscripted dialogue between the system and the performer, was improvised without any orientation except the performer's intuition. We analyze the movements, the occurrence of the expected events on the audio track, and any failures to assess the reliability of the system. Scenario 1 included discrete events, trigger mappings and gestures (Table 1).

Scenario 2, a public live performance by the acting class at our university, was given in two runs, of which the first is analyzed (see Section 4.2). It was choreographed as a planned sequence of movements that can be verified for correctness. The setup was additionally operated live rather than running autonomously: tracking was started manually when the performer entered the tracking area. Since it used two continuous mappings and only one trigger, we determine phase by phase whether the sound follows the parameter. In the chronological sequence of the phases there was flexibility regarding their exact timing, duration, and repetition.

Both trigger events and phases are coded as *correct* (the sound follows the movement or audibly changes the track), *delayed* (the offset is perceptible at normal playback speed), *missed* (a coded movement triggers no sound), *false trigger* (a sound occurs without a corresponding movement), or *uncertain* (not classifiable from the material).

The analysis is limited to the video and audio recordings and, for Scenario 1, a screen recording of the skeleton view in Kinect Studio (without numerical joint data). Continuous mappings are harder to pin to a point in time and are therefore coded less often than

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	Movement / Parameter	Type	Calculation / Condition	Sonic Role
Scenario 1	Body Height	continuous	`head.y − knee.y`	Intensity of drone layer
	Hand ↔ Hand Distance	continuous	euclidean distance (handR, handL)	Expands harmonic sound field and panning
	Hand Motion Speed	trigger	slope CHOP + trigger threshold	Triggers short sample events
	Foot Motion Speed	trigger	slope CHOP + trigger threshold	Triggers short sample events
	Hands in Front of Chest	gesture	relative hand position near the chest area	Activates an intimate sonic state
	Hands Above Head	gesture	both hands positioned above the head	Activates an expanded sonic state
Scenario 2	Body Height	continuous	`head.y − knee.y`	Auto-filter on the sea-noise layer: high frequencies are attenuated as the body lowers
	Hand Motion Speed *(l)*	trigger	slope CHOP + trigger threshold	Triggers a MIDI note (water-splash sample)
	Head ↔ Hand Distance *(r)*	continuous	euclidean distance (head, handR)	Heartbeat sound volume

Table 1: *Movement parameters and detected gestures in both scenarios, with their calculation and assigned sonic role. Scenario 2 uses a reduced set: three parameters and no gestures.*

their prominence in the sound suggests — a bias that tends to work against the system.

4.2. Reliability in Deployment

Scenario 1 — observed failure types. For Scenario 1, a duration of 3:44 was analyzed from the six-minute recording, resulting in 78 recorded events, which achieved sufficient saturation. Of the 78 events, 46 were marked as *correct*. An analysis of the false events revealed 20 false triggers compared to only 3 missed events. In most cases, the skeleton view shows that the Kinect's tracking itself was responsible — the overlapping of joints described in Section 3.3 (see Figure 5; 15/20 false triggers, 3/4 delays): during small steps or while balancing on one leg, the skeleton fitting faltered, and the resulting uncontrolled bouncing of the joints in the extremities crossed the trigger thresholds or disrupted gesture recognition. Two further false triggers were caused by a location-specific condition (a railing blocked the movement).

Scenario 2 — phase-by-phase correspondence. The recording of Scenario 2 was divided chronologically into eight phases; the system performed successfully in six of them. The body height tracking followed the performer's posture continuously and without perceptible latency (Phases 1, 2, 4, 7, 8). The head-hand tracking made the heartbeat sound audible as soon as the hand was moved toward the head (3, 5, 7), and the only trigger mapping responded reliably (6/6; Phase 6). Two phases deviated: in Phase 1, hand movements at the level of his torso were unintentionally detected and led to a *false trigger*; in Phase 3, the recording gives the impression of slight latency, which could be attributed to the intervals within the audio track.

System availability. During Run 2 of the public live performance, the system was prevented from starting due to an irreversible total failure of the sensor, presumably caused by a short circuit while the components of the pipeline were being reconnected.

Can the system be considered reliable? During operation, the system was reliable enough for the dialogue examined in Section 4.3; its availability, however, was not ensured, as Run 2 showed (no second Kinect as a backup). The errors in Scenario 1 were mostly additional sounds rather than gaps (20 false triggers vs. 3 missed events), and in Scenario 2, six of eight phases followed the movement, as did all six intended triggers. In both scenarios, the system was therefore oversensitive rather than unresponsive: these deviations did not interrupt the connection between the performer and the system.

4.3. Movement Vocabulary

In a questionnaire, the performer answered 10 open-ended questions about his interaction with the system, covering movement–sound relationships, the development of the solo sequences, the limitations of the system, and the evolution of his body awareness over the course of the sessions. The performer's feedback shows why the reliability matters for the dialogue: when the system did not respond, it felt to him like a conversation partner who stops answering; when it did, listening to the sound became part of his performance. This leads to the question: *How did the system shape the performer's behavior and movement sequences?*

Whereas in Scenario 1 the sound mapping was assigned by by the system's developer, the performer chose his choreography for the public live performance (Scenario 2) by himself. He came up with a concept for this scene, which was thematically integrated into the narrative of the performance, but its precise execution was shaped by the system's limitations, as mentioned in Section 3.3: facing the camera head-on, no kneeling, and no turns. During the test procedures, the performer developed a movement vocabulary that the Kinect reliably registers, while, according to his own statement, he deliberately avoided problematic sequences. His selection criteria evolved toward clear, repeatable, and rather mechanical movements, and away from movements through space. A reciprocal process had already emerged during the development process: "Everything built on one another [...] after the initial tests, new ideas came up, which I incorporated. The sound then changed in turn, following those new ideas" (translated from German). Furthermore, he notes that the project felt freer than his usual rehearsal process, even though the freedom of movement was much more limited. The freedom lay much more in not knowing the final outcome — what he had originally planned could not be carried out, but the system opened up a room for experimentation: "The sounds and the limitations and possibilities of the system meant that some of what was in my head would not have worked and, in my opinion, that led to much better ideas" (translated from German).

This dialogue can also be observed in the recording from Scenario 2: In Phase 6, the left hand triggers a wave sample. At first,

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this stands out only slightly against the background sound. The performer notices this and adjusts the intensity of his movements accordingly. Later, he says that, paradoxically, it was this precision that allowed him to convey more effectively to the audience what he originally intended to communicate. He also reported on the effect of relational body mapping that "the system is more reliable in terms of how I perceive my own movements, since that is also my frame of reference" (translated from German).

5. Discussion

Taken together, the two scenarios suggest that the constraints of the low-cost setup shaped the performer's movement vocabulary. The observations have to be weighed against explanations that do not involve the system at all.

The practice was just as important as the technology: In addition to the hypothesis that relational data improved the mapping, the performer became more proficient in using the system with each use. Furthermore, the system did not run completely autonomously and was accompanied live, though none of these interventions constituted error correction. These explanations mean that the changes from Scenario 1 to Scenario 2 cannot be attributed to the system alone. The reciprocal dialogue described in Section 4.3, however, does not rest on that comparison: in Phase 6 the adjustment is observable within a single performance, independently of the performer's own account. Reduction was also identified as a distinct finding. The iterative revisions ultimately led to a more streamlined mapping. Scenario 2 ran on a much smaller system than Scenario 1, so the two scenarios cannot be compared based on an identical mapping system. Nevertheless, the performer later mentioned that he even liked these constraints.

The chosen movements also determined how fine the resolution of the mapping could be, given the limited scope of body-centric data that Schacher [Sch10] describes (Section 2). In Scenario 2, both hands operated within a small radius close to the head, which had two separate consequences. First, trigger points had to be set low, because the short distance of the movement results in a lower measured hand speed. Intended movements were therefore hardly distinguishable from unintended ones, which resulted in a conflict when adjusting the threshold: a low threshold leads to false triggers, such as the hand movements at torso level in Phase 1, and a high threshold leads to a lack of feedback. In Scenario 2, the threshold was therefore kept low, and false triggers had to be accepted in favor of reliable feedback. Second, the short distance led to an abrupt change in sound during the continuous hand–head mapping, since only a small range of movement was available for sound development: the value crossed its entire range within a brief gesture, which may explain the impression of latency in Phase 3.

The clearest limitation, however, was not accuracy but availability: since no second sensor was available, the total failure in Run 2 (Section 4.2) could not be absorbed. The risk lies less in the inexpensive hardware itself than in how it is assembled: a low-cost pipeline chains separate components, each connection is a potential point of failure, and without redundancy, a single failure stops the entire system. Low-cost setups are therefore not risk-free. Further limitations apply: a commercially licensed audio environment (Section 3.1), thresholds tuned to one single body (Section 3.5), and a single performer who co-developed the system (Section 4.1). While these limitations can largely be attributed to this specific pipeline, the body as a design principle is, in contrast, a transferable concept that can be applied to other sensor technologies beyond this pipeline.

6. Future Work

Future work could address these limitations for further exploration: the system could be tested with multiple performers, and their perceptions could be incorporated alongside the data from the questionnaire. The performer also suggested designing the system to accommodate multiple people at the same time, assigning sounds to distinguishable individuals, and making greater use of the available space (combining absolute and relational mappings). Furthermore, Ableton Live could be replaced with a non-commercial DAW, or sounds could be controlled within TouchDesigner alone. In addition, the audience's perception could be taken into account to highlight the contrast between measurable events and the actual effect.

7. Conclusion

Based on the analysis, we documented that the limitations of low-cost motion tracking within a modular pipeline became a guiding principle and helped shape the movement vocabulary. In the development tests, body-relative mapping yielded control values that are observed as more stable, while the available range of movement limited the resolution of the mapping. The system presented here operates under restrictive conditions and is not error-free, but it establishes a dialogue between the body and technology. The performer experienced the process as freer, even though his range of movement was more limited, because its outcome remained open. Thus, limits of low-cost motion tracking need not be overcome before artistic work can begin; they can be explored during the process and become part of the final creative result.

Acknowledgements

Thanks to all the fellow students involved in the project, with special thanks to the performer for his collaboration and valuable feedback throughout the studio sessions. Thanks also to our professors for their guidance and support.

We acknowledge the use of the AI tools Claude Opus 5.5 (Anthropic) and DeepL (web version, 2026). Claude was used selectively for refining and condensing specific passages, for sharpening arguments and technical terminology through iterative rephrasing, for grammar correction and language polishing, and for creating the diagrams. DeepL was used for translation throughout the work. The author is solely responsible for all content and claims.

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